SIX1015 Science, Technology and Society

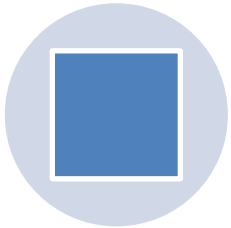
Semester 2 2025/26



LECTURE 1: INTRODUCTION TO SCIENCE, TECHNOLOGY & SOCIETY

Dr Nurulaini Abu Shamsi Abu Shamsi

Lecture outline



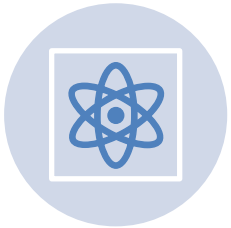
COURSE INFO & PROFORMA



DEFINITION



BRIEF HISTORY OF SCIENCE



WHY STUDY SCIENCE, TECHNOLOGY AND SOCIETY?

Ice-breaking







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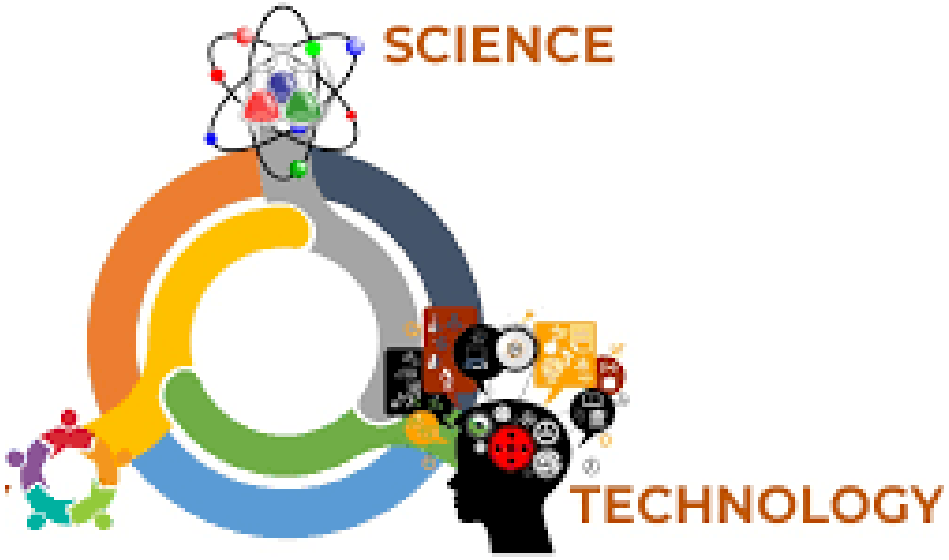
A scientist working in a lab

- Who funds this research?
- Why is this research done?
- Who benefits?
- Can science ever be neutral?



WHAT'S BEYOND LAB?

PASSWORD: Awesome Possum!



SCIENCE, TECHNOLOGY AND SOCIETY
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SCIENCE & TECHNOLOGY STUDIES



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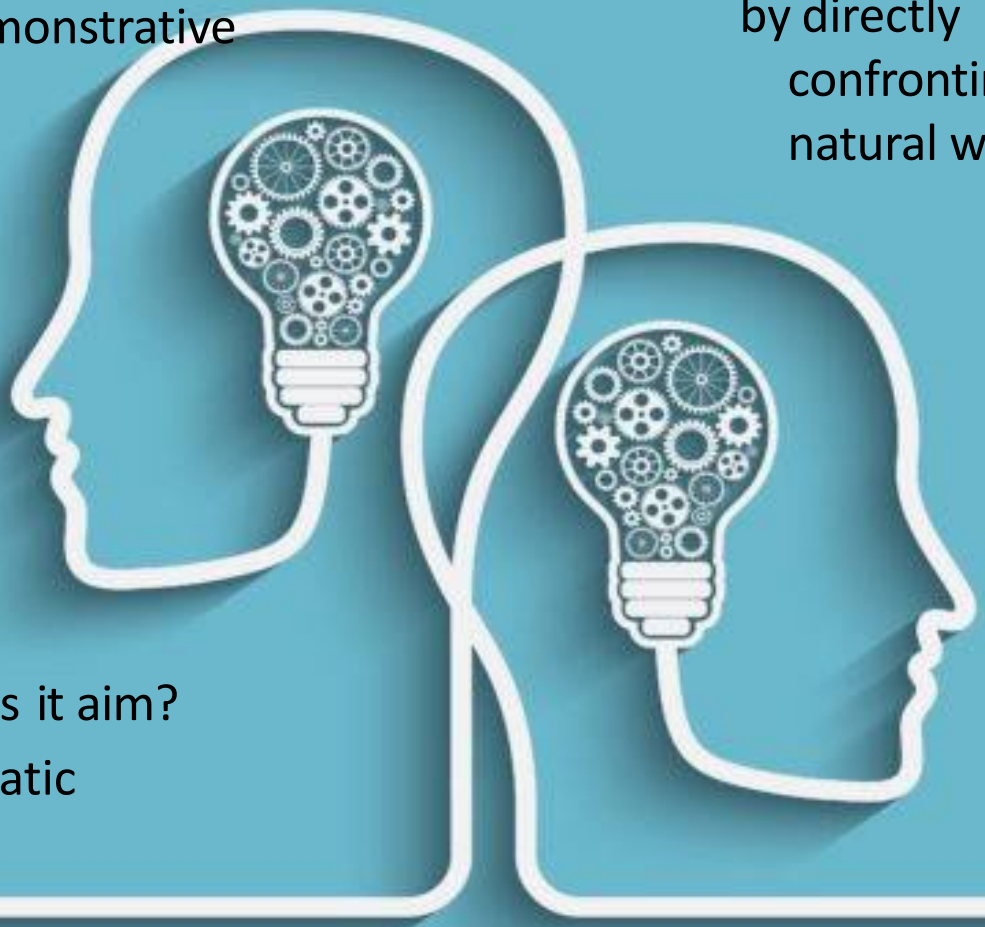
DEFINITION



SCIENCE

Science comes from the Latin "scio" meaning “I know.” Scio derives from the Latin infinitive "scire" meaning "to know.“ “Scientia” in Latin means “knowledge” – “demonstrative knowledge.”

Human activity that creates and accumulates knowledge by directly confronting the natural world.



What is the aim of science?

- To know the truth about the world

How science achieves it aim?

- By using a systematic scientific method



- Science is concerned with evidence and with theory.
- Scientific evidence often comes from experiments
- To explain the evidence, theories are put forward, and further evidence is often sought, to see whether the theory accords with additional observations.

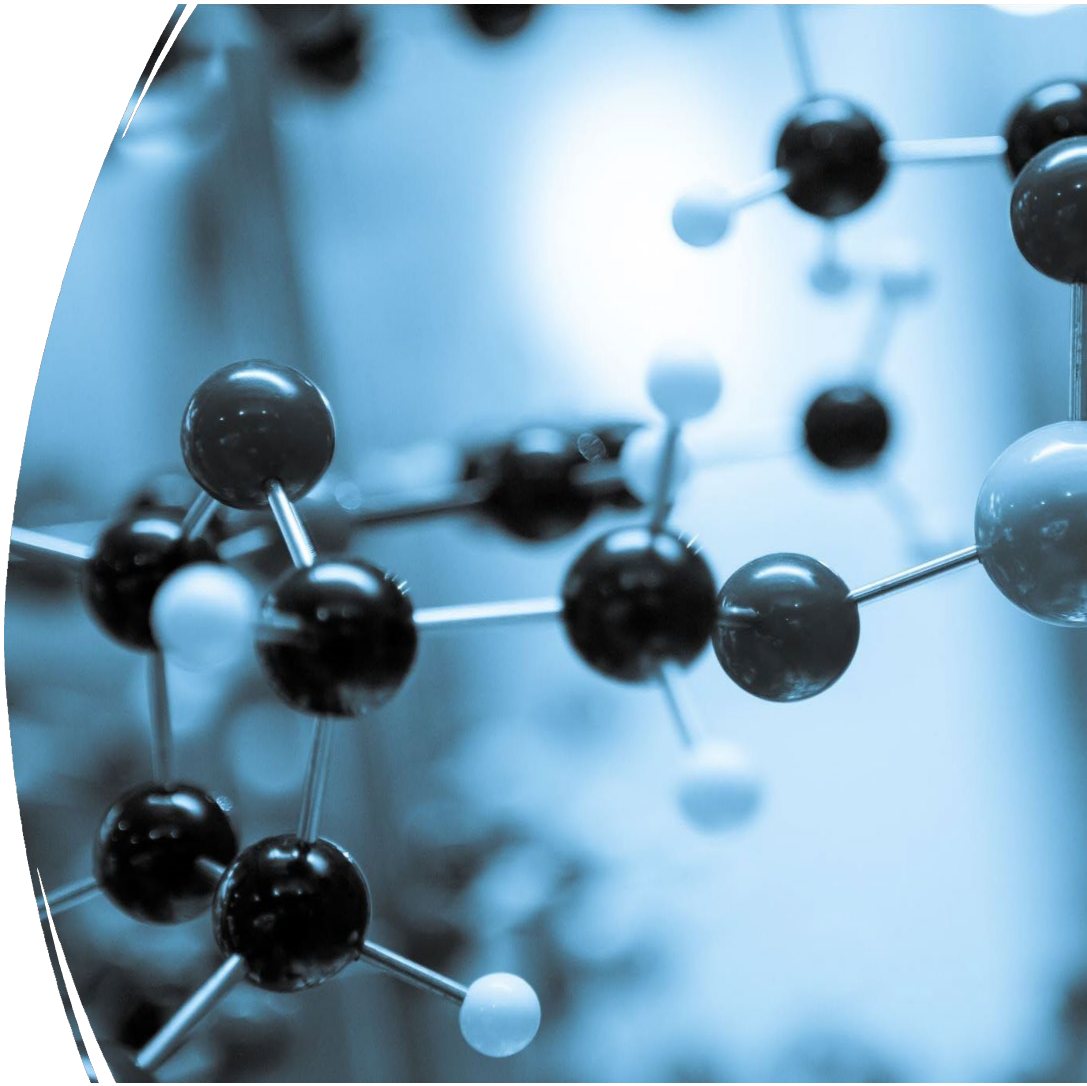
HOW DO WE KNOW THAT SCIENTIFIC KNOWLEDGE IS RELIABLE?

Why do we trust science more than other forms of knowledge like beliefs, traditions, or intuition?

WHAT IS PHILOSOPHY OF SCIENCE?

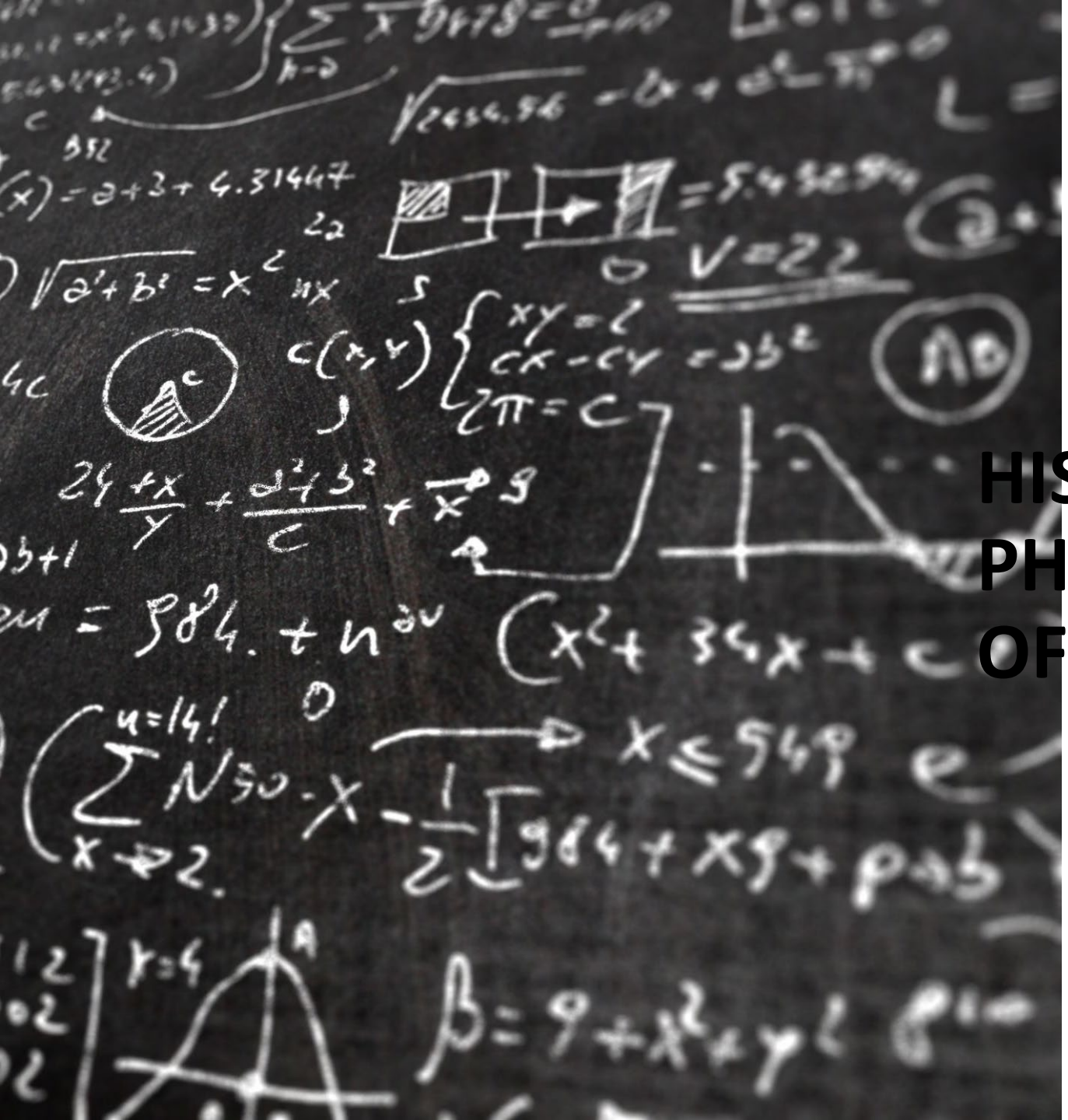
- Philosophy of science is a branch of philosophy that attempts to **elucidate** the nature of scientific inquiry and **evaluate the grounds of their validity**. It is the study of assumptions, foundations, and the **implications of science**.

- The field deals with what science is, **how it works**, and **the logic** through which we build scientific knowledge.



Philosophy of science asks questions like:

- What counts as scientific knowledge?
- How do scientists know something is true?
- What makes science different from other knowledge?
- Can scientific knowledge ever be completely certain?



HISTORY AND PHILOSOPHY OF SCIENCE

A Timeline of Scientific Milestones

Ancient Civilizations

Ancient Greece 3000 BCE - 500 CE)

Islamic Golden Age (7th-15th) centuries

European Renaissance and Scientific Revolution:

Period	Civilization
Ancient Period (3000 BCE - 500 CE)	Ancient Egypt Ancient Mesopotamia
Medieval Period (500 CE - 1500 CE)	Ancient Greek

Early Modern Period (1500 CE - 1789 CE)	Islamic Golden Age	Astronomy, mathematics, medicine, chemistry, physics
	Scientific Revolution	Astronomy, physics, chemistry, biology
Modern Period (1789 CE - Present)	Industrial Revolution	Physics, chemistry, engineering
	Modern Physics	Quantum mechanics, relativity
	Modern Biology	Genetics, evolution, molecular biology
	Information Age	Computer science, information technology

When did modern science begin?

Science in earlier civilizations

Translations and transformation

Early modern science in Western Europe





The history of science from the ancient Greeks to the present is the history of one compartment of philosophy after another breaking away from philosophy and emerging as a separate discipline.

Ancient Greeks – Philosophy and science are not separated.
Euclid's work had made geometry a "science of space" separate from but still taught by philosophers in Plato's Academy.

17th century
Galileo, Kepler and finally Newton's revolution made physics a subject separate from metaphysics.

1859
"The Origin of Species" set biology apart from philosophy

20th century
psychology broke free from philosophy

logic has given rise to computer science.

When did modern science begin?

What made it possible for science to acquire prestige and power in Western Europe by the 17th century?

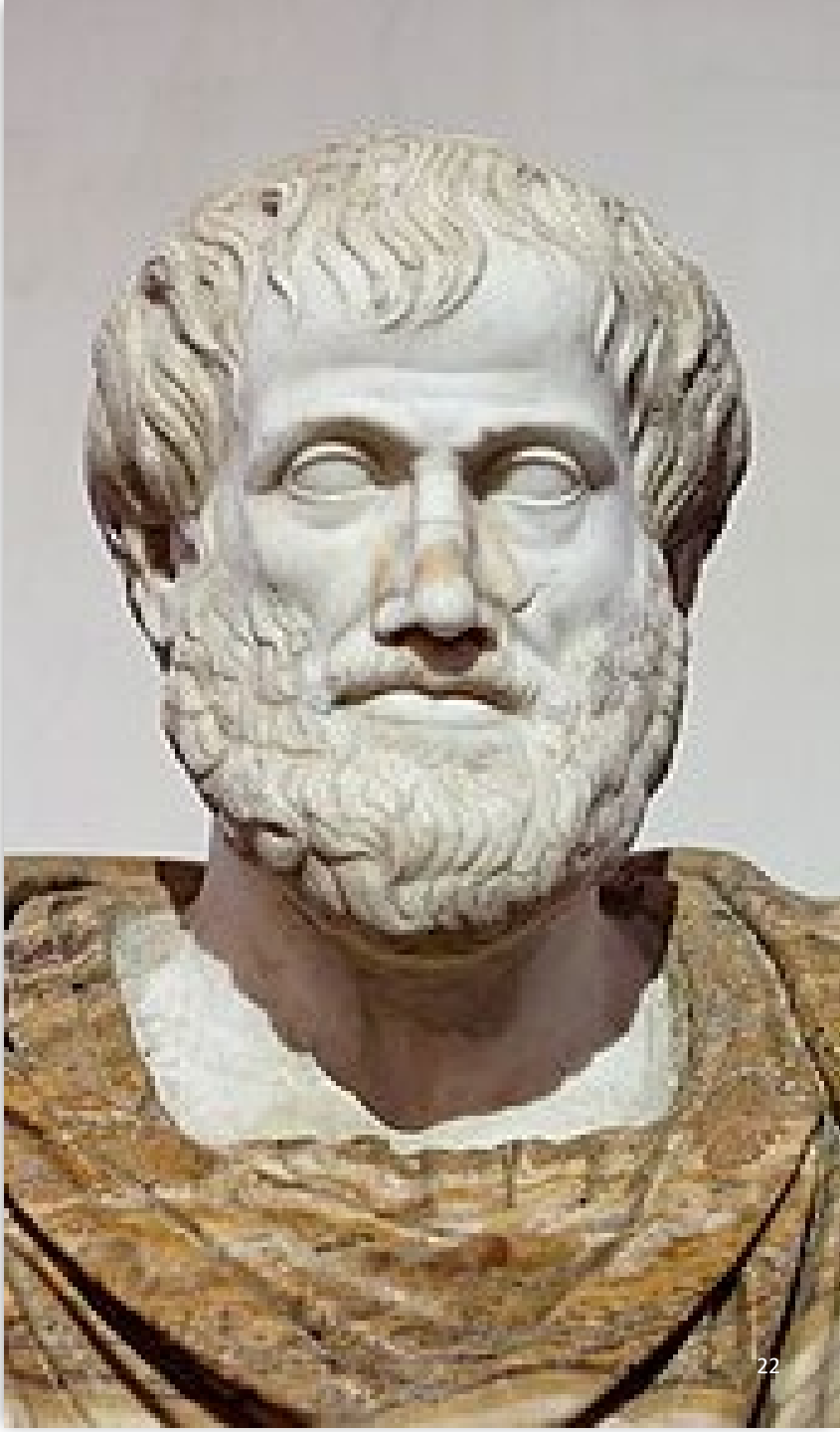
- Changes that created a conducive environment for the birth of science
- Translation of Greco-Arabic science and natural philosophy into Latin
- Establishment of medieval universities
- Emergence of a class of theologian-natural philosophers that encouraged the use of Aristotelian natural philosophy in the curriculum of universities
- Medieval natural philosophers produced hundreds of questions about nature, the answers to which included a vast amount of scientific information.

The scientific revolution and the new tool of induction

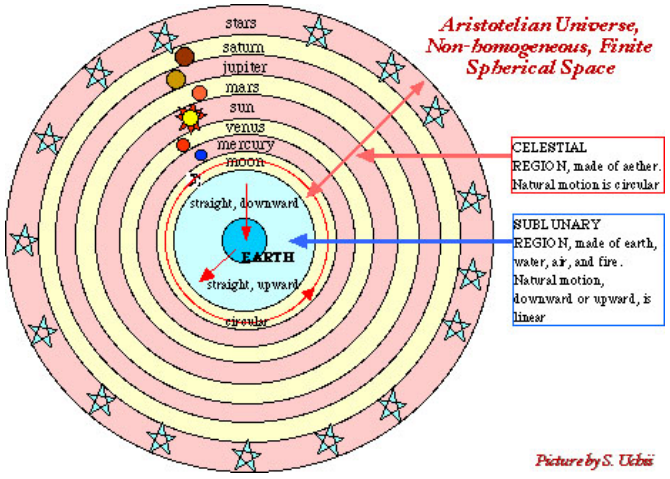
- Most of the questions had multiple answers, with no genuine way of choosing between them.
- In 16th and 17th century, new answers were proposed by scholars who found Aristotelian answers unacceptable or inadequate.
- From philosophical point of view, the most important development during the Scientific Revolution was the increasingly widespread break with the theories of Aristotle.

Event 1: Heliocentric vs geocentric

- Aristotelian view was the dominant philosophical view in the Western world prior to the scientific revolution.
- Aristotle's philosophy had been combined with the **doctrines of Christianity to form a cosmology and philosophy of nature (Scholasticisim)**

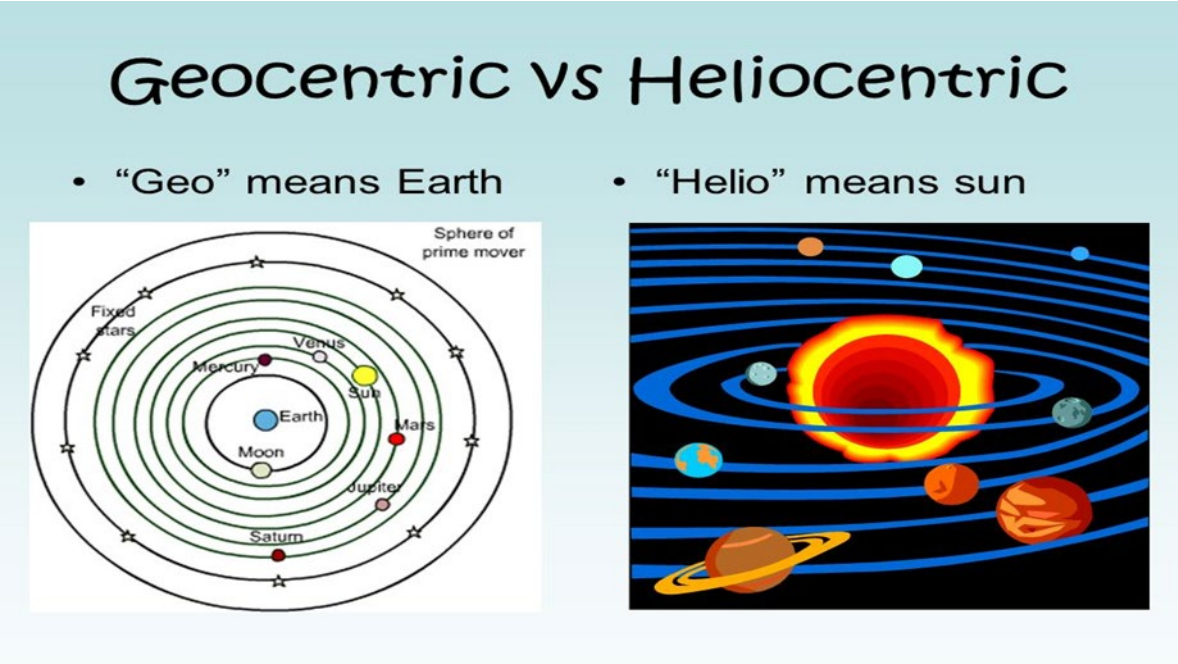


- **Scholasticisim –** described everything from the motions of the planets to the behavior of falling bodies on the earth, the essential of which were largely unquestioned by most western intellectuals.
- Aristotelian view –
  - Geocentric Model: Aristotle believed the Earth was the center of the universe (geocentrism), a view later challenged by Copernicus and Galileo.
  - Four Elements Theory: He proposed that all matter is made up of four elements: earth, water, air, and fire.
  - Teleology (Purpose in Nature): He argued that everything in nature has a purpose or end goal (telos).



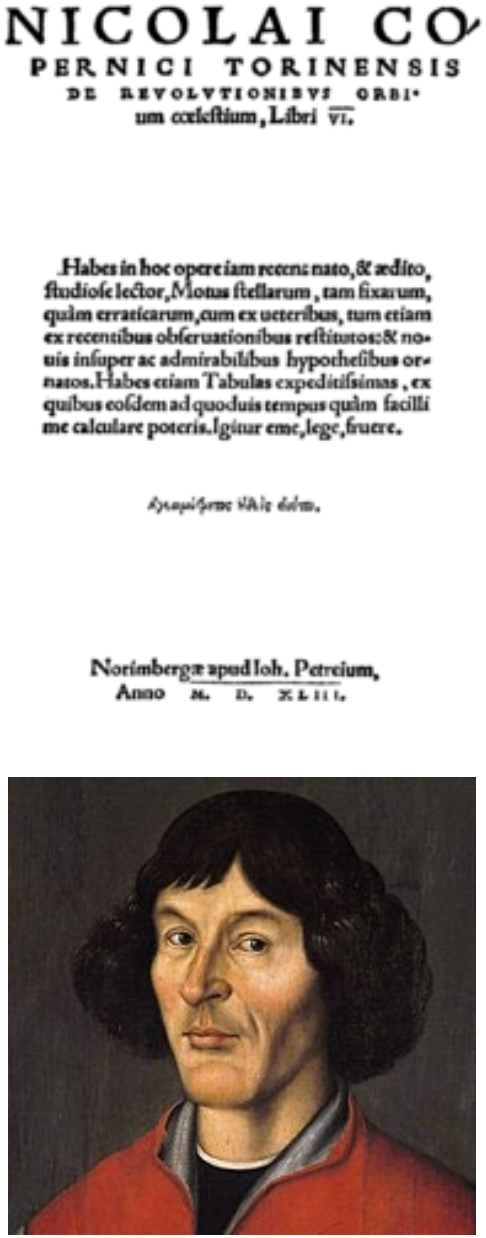
COPERNICAN REVOLUTION

- Copernican revolution: a shift in beliefs about the solar system and the wider cosmos, from geocentrism to heliocentrism



Heliocentric vs geocentric

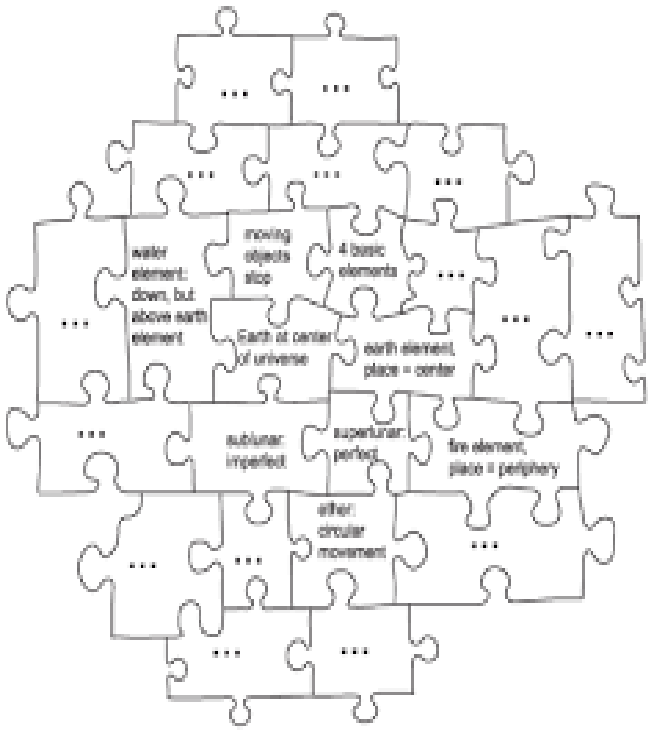
- In 1542, the astronomer Nicolas Copernicus (1473-1543) published a book attacking the geocentric model of universe, which placed the stationary earth at the centre of the universe with the planets and the sun in orbit around it.
- Geocentric astronomy, also known as Ptolemaic astronomy after the ancient Greek astronomer Ptolemy, lay at the heart of the Aristotelian world-view, and had gone largely unchallenged for 1,800 years.


NICOLAI CO
PERNICI TORINENSIS
DE REVOLVTIONIBVS ORBI
um coelestium, Libri VI.

Habes in hoc opere iam recens nato, & ædito,
studiose lector, Motus stellarum, tam fixarum,
quàm erraticarum, cum ex ueteribus, tum etiam
ex recentibus obseruationibus restitutos: & no-
uis insuper ac admirabilibus hypothesibus or-
natos. Habes etiam Tabulas expeditissimas, ex
quibus eosdem ad quoduis tempus quàm facilli
me calculare poteris. Igitur eme, lege, fruere.

Norimbergæ apud Ioh. Petreium,
Anno M. D. XLIII.

- But Copernicus suggested an alternative: the sun was the fixed centre of the universe, and the planets, including the earth, were in orbit around the sun.
- On this heliocentric model the earth is regarded as just another planet, and so loses the unique status that tradition had recorded it.



- Copernicus' theory initially met with much resistance, not least from the Catholic Church who regarded it as contravening the Scriptures and in 1616 banned books advocating the earth's motion.
- But within 100 years Copernicanism had become established scientific orthodoxy.
- Copernicus' innovation did not merely lead to a better astronomy.



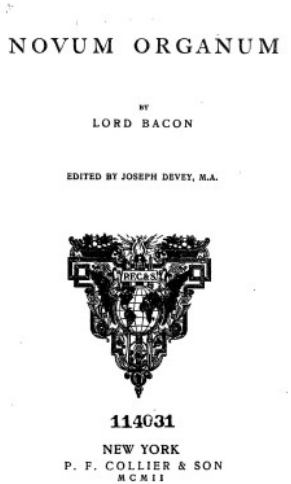
The origins of modern science

- Indirectly, it led to the development of modern physics, through the work of Johannes Kepler (1571-1630) and Galileo Galilei (1564-1642).
- Kepler discovered that the planets do not move in circular orbits around the sun, as Copernicus thought, but rather in ellipses. This was his crucial ‘first law’ of planetary motion; his second and third laws specify the speeds at which the planets orbit the sun.



Event 2: **Francis Bacon (1561–1626) revolutionized science** by introducing a new empirical approach to knowledge.

- Among the propagandists was Francis Bacon (1561-1626) who proposed a new method for the sciences to replace that of Aristotle.
- He published Novum Organum to replace Aristotle's Organon.
- Baconian method is based on two pillars: observation and induction.
- Observation: without prejudice or preconception, write observation statements i.e record the results of what we see, hear, smell, of the world as we find it, or of the special circumstances of our experiments. The more data, closer to the truth.
- Induction: Based on the observation, universal generalisations are made.


NOVUM ORGANUM

BY
LORD BACON

EDITED BY JOSEPH DEVEY, M.A.

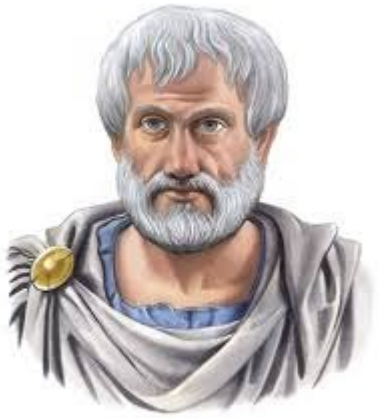
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- **The Baconian Method:**
  - Rejected Aristotle's **deductive reasoning**, which relied on abstract logic rather than real-world evidence.
  - Advocated for **inductive reasoning**, where scientists gather data through observation and experiments before forming theories.
  - Published *Novum Organum* (1620), laying the foundation for the **scientific method** still used today.

Deductive logic / Syllogism (Aristotl



All human beings are mortal (premise 1)

Socrates is a human being (premise 2)

Therefore Socrates is mortal (conclusion)

What thinks must exist

I am thinking

Therefore_____ (cogito ergo sum)

Problem:

- the argument does not expand our knowledge because the conclusion reveals what premises already state.
- Little role of actual sense experience in the acquisition of knowledge

- Problem of induction:

- An inductive argument could always be false no matter how many observations we have made.

Eg: All swans are white; Europeans believed this but Australia has black swans.

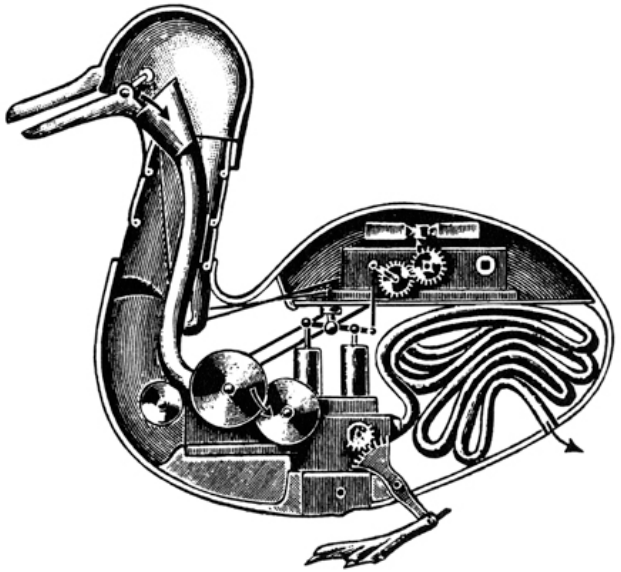
- We also cannot make any observations without some ideas concerning the nature of what it is we are observing.



Event 3

- The French philosopher, mathematician and scientist Rene Descartes (1596-1650) developed a radical new mechanical philosophy, according to which the physical world consists simply of inert particles of matter interacting and colliding with one another.



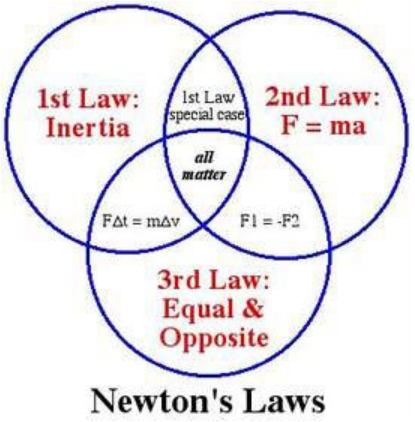


An automaton of a duck, inspired by the mechanical philosophy, created by Jacques de Vaucanson in 1739. René Descartes had written, "I have described this earth, and indeed this whole visible world, as a machine," and under the spell of the mechanistic philosophy, even animals and the human body came to be pictured as machines rather than living organisms. (source)

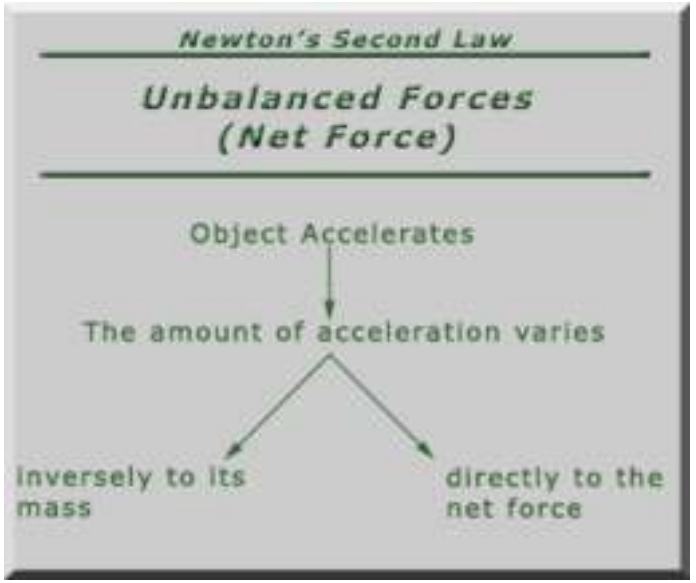
Event 4

- Isaac Newton's (1643-1727) masterpiece was his Mathematical Principles of Natural Philosophy, published in 1687, improved Descartes' laws of motion and rules of collision. Newton's three laws of motion and principle of universal gravitation.
- Newtonian physics provided the framework for science for the next 200 years. The 18th and 19th centuries both saw notable scientific advances, particularly in the study of chemistry, optics, energy, thermodynamics, and electromagnetism.





Newton's Laws

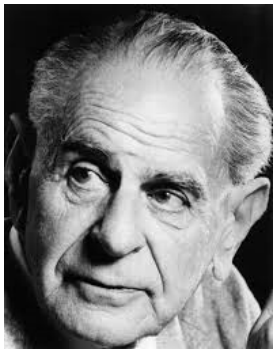


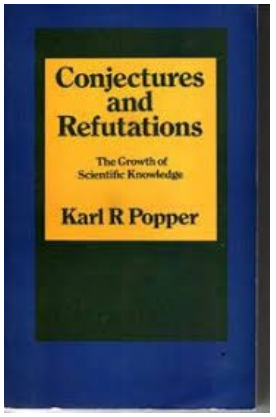
Karl Popper and the Problem of Induction

Karl Popper (1902–1994)

Key idea:

- Science progresses through **testing and falsifying theories**
- Problem of induction:
- No number of observations can prove a theory true
- A single observation can **disprove it**
- Example:
  All swans are white → Black swan discovered
- **Scientific theories must be falsifiable**





Popper: Science vs Pseudoscience

Popper's **criterion of falsifiability**

- A theory is scientific if it:
  - Makes **testable predictions**
  - Can potentially be **proven wrong**
  - Scientific theories:
  - Accepted **temporarily**
  - Open to **revision**
- Examples of scientific change:
    - Ptolemy → Copernicus
    - Newton → Einstein
- Scientific change = **progress toward better explanations**

Thomas Kuhn (1922–1996)

- Kuhn studied the **history of science**
- Key argument:
  Science does **not always progress gradually**
- Instead, science develops through:
- Long periods of **normal science**
- Occasional **scientific revolutions**
- Main idea:
  The Structure of Scientific Revolutions

Paradigm and Normal Science

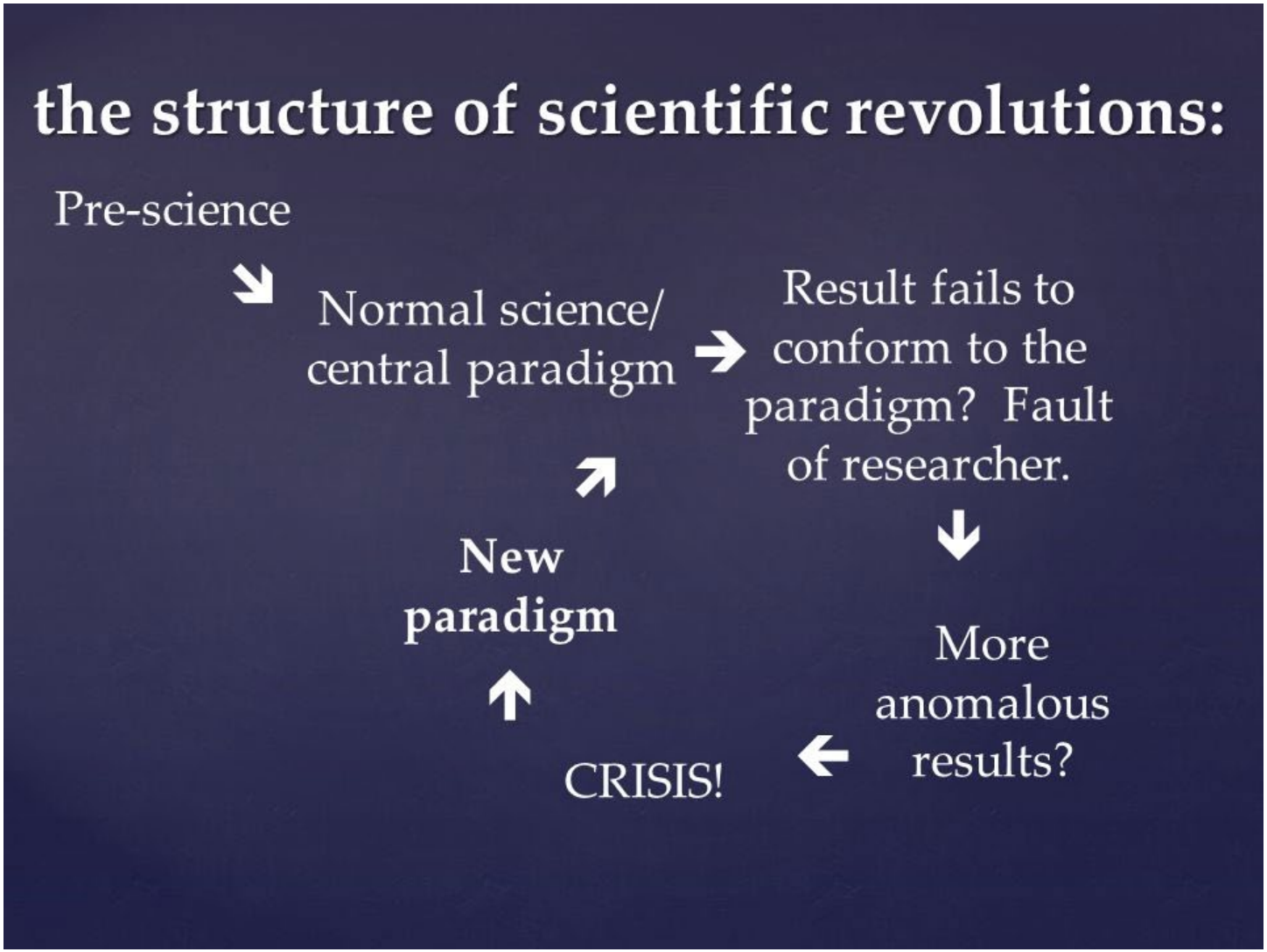
- **Paradigm:** A shared framework that guides scientific research
- Includes:
  - Accepted theories
  - Research methods
- Key examples of scientific work
- During **normal science**:
  - Scientists solve problems within the paradigm
  - Research is structured and organized
  - Kuhn describes this as **puzzle-solving**

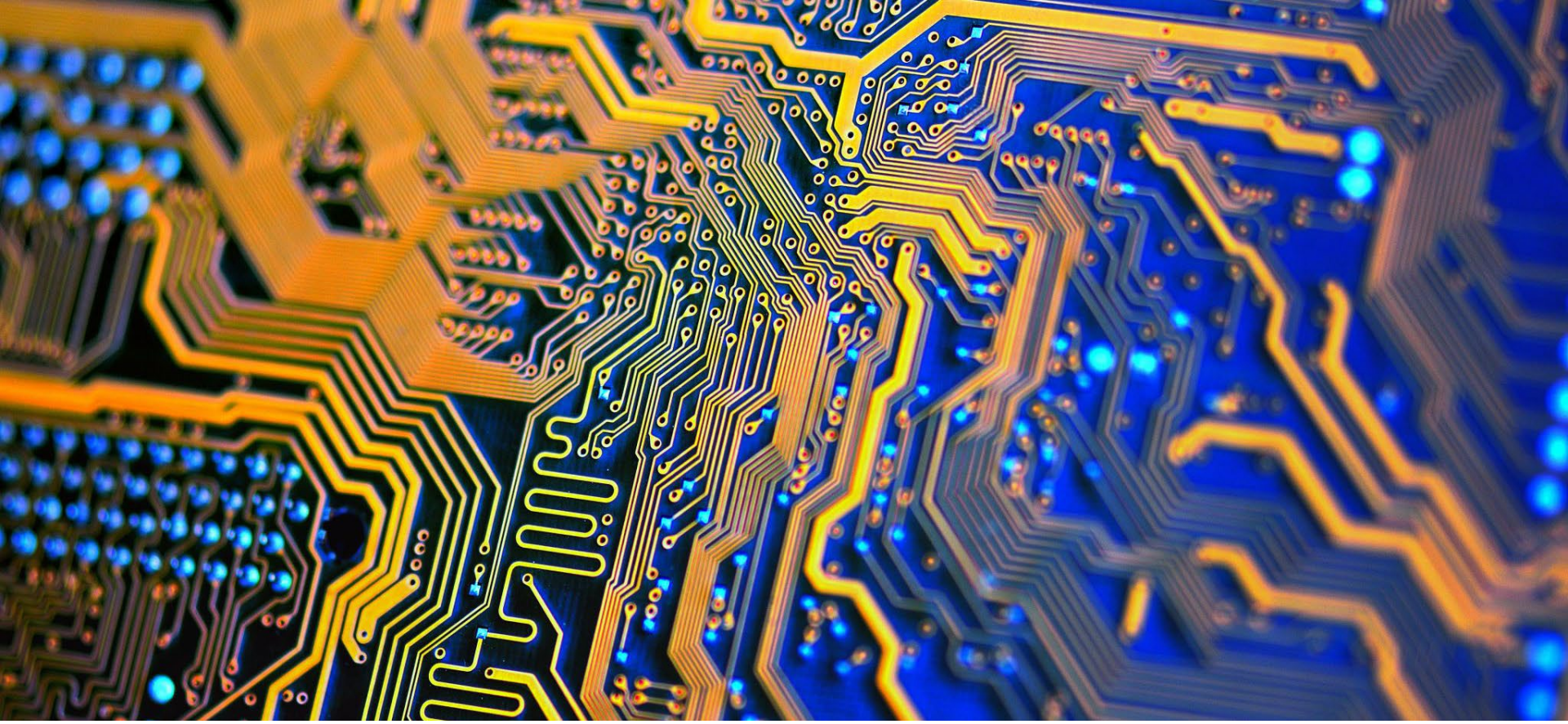
Anomalies and Crisis

- Sometimes scientists encounter results that **cannot be explained**.
- These are called **anomalies**.
- At first anomalies are ignored, but if they accumulate:
- Normal science → **Crisis**
- Scientists begin searching for **new explanations**

Scientific Revolution

- When a new framework replaces the old one: **Scientific Revolution**
- Example:
  Geocentric model → Heliocentric model
- Old paradigm replaced by a **new paradigm**
- Science progresses through **cycles of change**

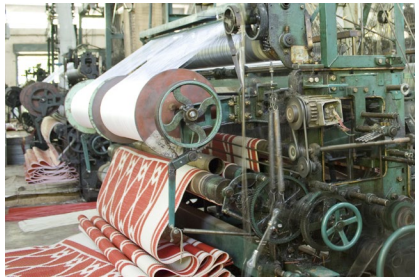

the structure of scientific revolutions:
Pre-science
Normal science/ central paradigm
Result fails to conform to the paradigm? Fault of researcher.
More anomalous results?
CRISIS!
New paradigm

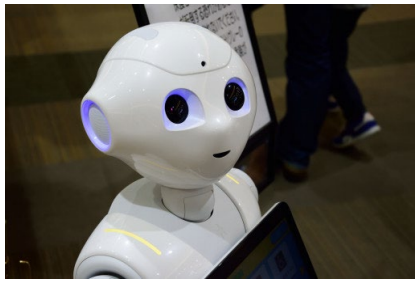
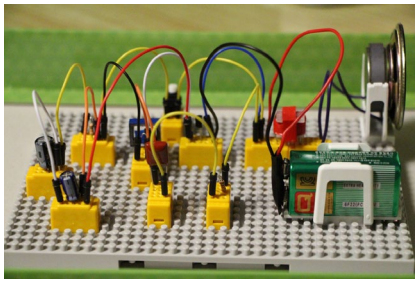


TECHNOLOGY

Technology

- Greek word “techne” = "craft, art, skill or the way a thing is gained/ tool“; “logos”= “word, thought”
- Application of knowledge for practical purposes, to make things and achieve humanly useful results.
- In a broad sense, technology includes the use of fire and stone axes and thus antedates science.
- Unlike science, moreover, technology is common in some form to every human culture that has ever existed.



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- Technology is not necessarily an application of science.
  - It can be based on craft rather than science, with knowledge being slowly accumulated and applied (often through trial and error) and passed on from one generation to the next. E.g irrigation systems
- More and more, technology is being influenced by scientific knowledge, with spectacular results.-R&D. They are all carried out in social, political and economic contexts.



SOCIETY



Society

Some definitions of society:

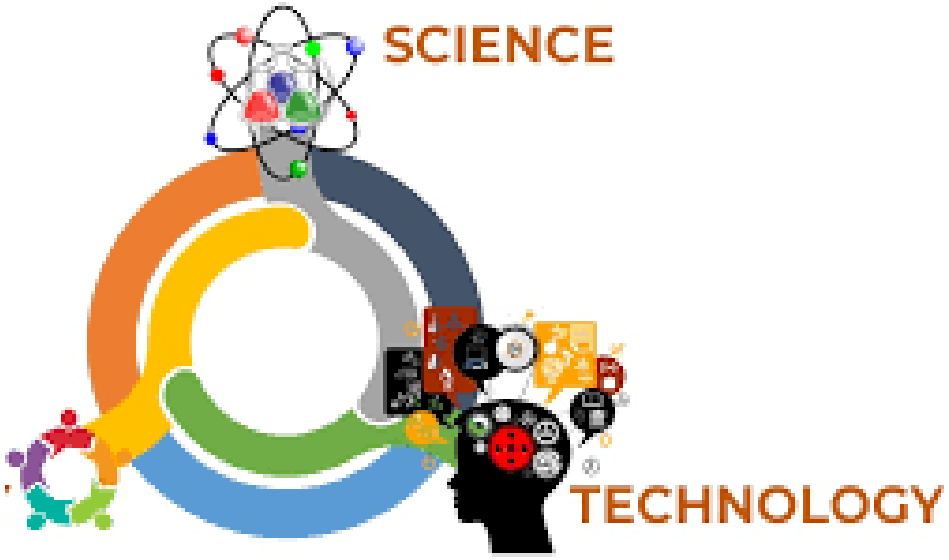
- an organized group of people associated as members of a community
- an organized group of persons associated together for religious, cultural, scientific, political, patriotic, or other purposes.
- a highly structured system of human organization for large-scale community living that normally furnishes protection, continuity, security, and a national identity for its members. Such a system is characterized by its dominant economic class or form: *middle-class society; industrial society.*

- *Society* can be applied to science and technology. Scientists and technologists do not work in isolation. They work in universities, firms or research groups.
- Questions about ethics and conduct within these groupings, how they should be financed and how they are best organised, are also matters which researchers in Science, Technology and Society can ask about.







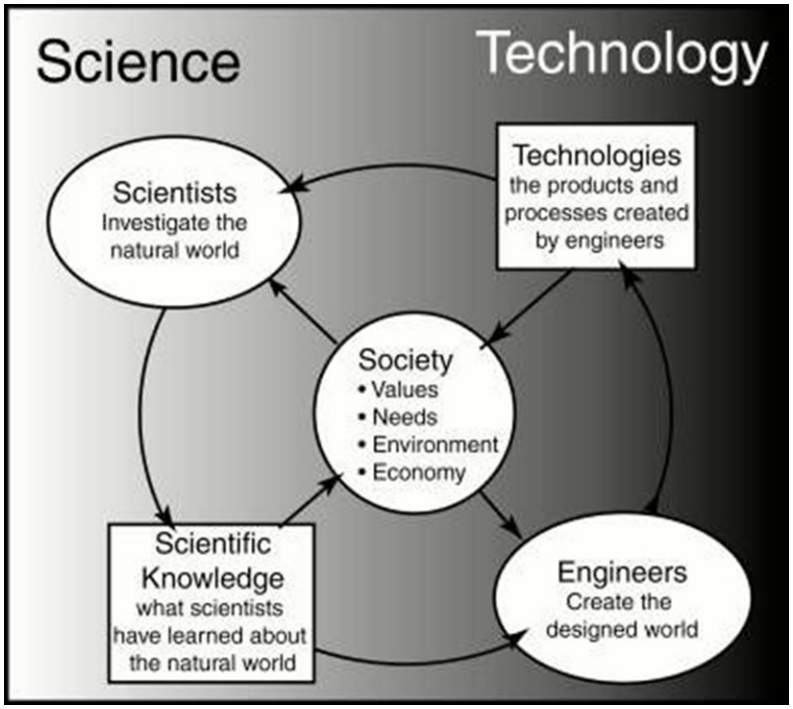


WHY STUDY SCIENCE, TECHNOLOGY AND SOCIETY?

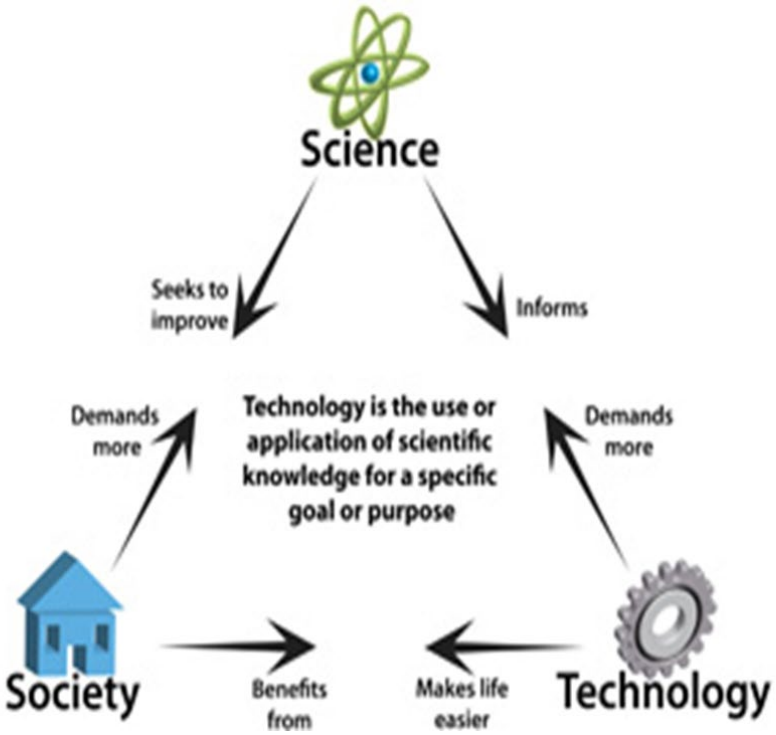
Science and Technology Studies

- STS draws from various disciplines, including history, sociology, philosophy, political science, and cultural studies, to understand the complexities of science and technology.
- STS emphasizes that scientific knowledge and technological advancements are not solely determined by nature but are also influenced by social, political, economic, and cultural factors.
- STS investigates how science and technology shape society and how, in turn, society influences the development and application of science and technology.

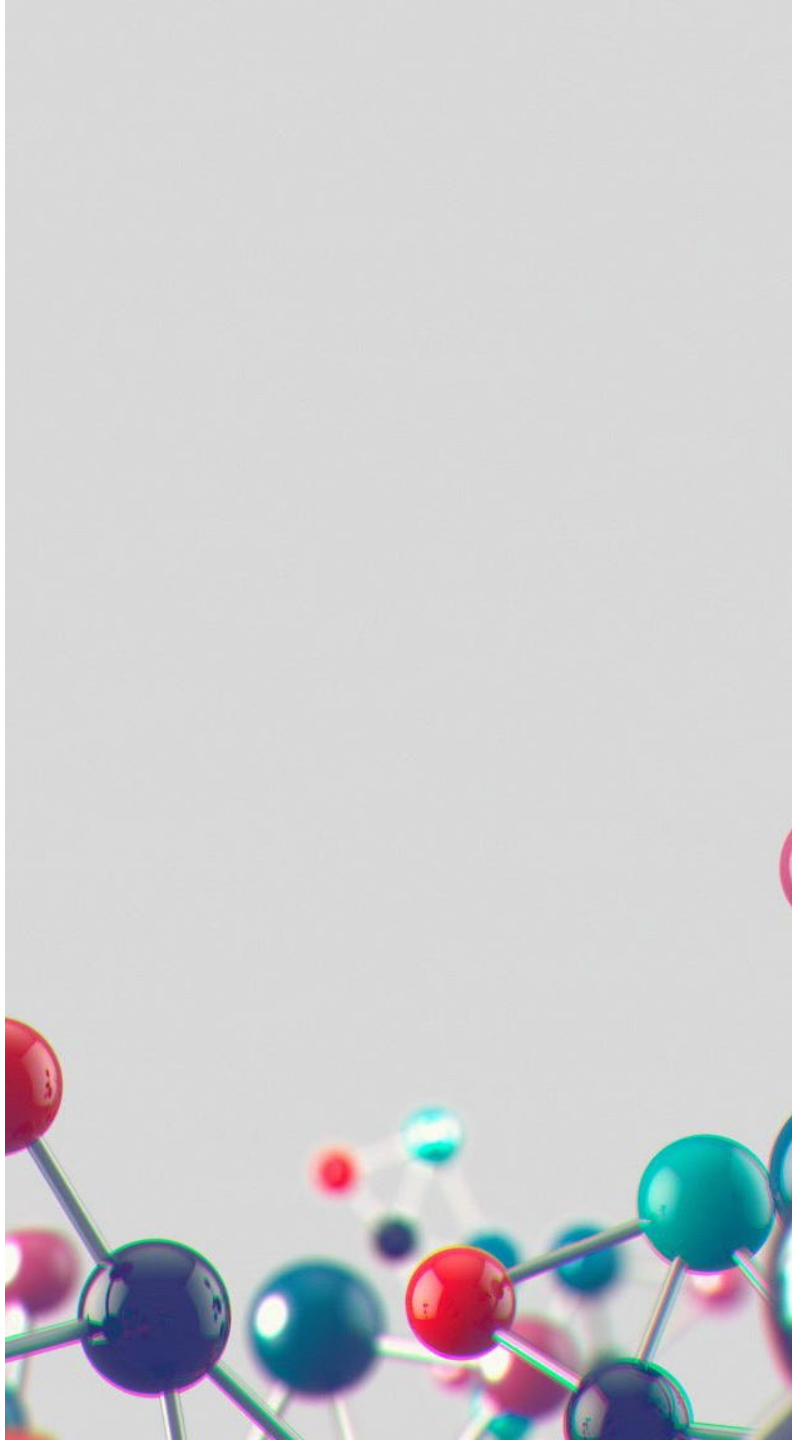
The Relationship between Science, Technology and Society



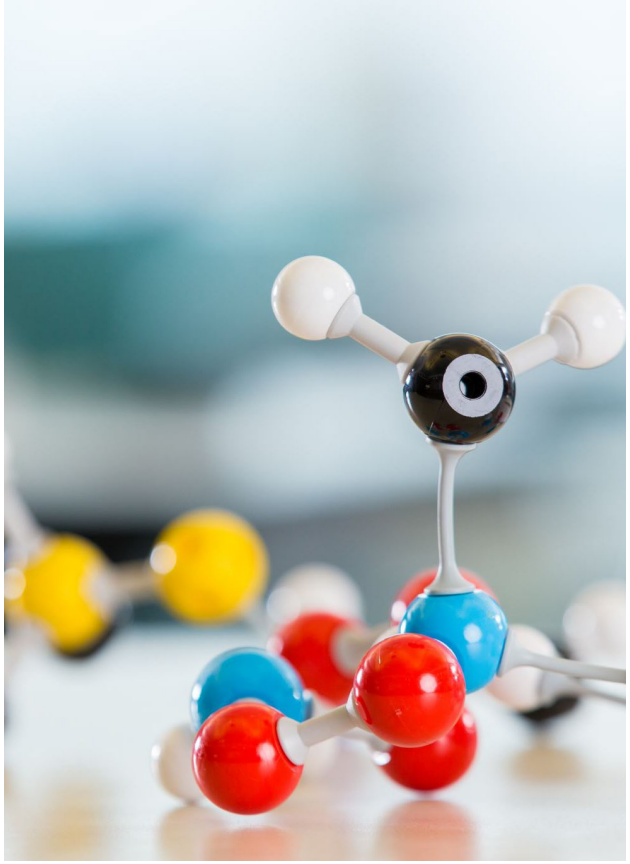
- The science understanding of how the world functions alters how we behave.
- Technology alters how we can behave.
- Society drives technological innovations and fosters scientific inquiry.
- Science gives us insight into the kinds of technologies we could potentially create and how to develop them, while technology enables us to conduct further scientific research.



- The fact that science is always the product of human activity.
- We are all familiar with terms such as the 'progress of science' or the 'onward march of technology', and we tend to forget that all scientific knowledge has been produced by people thinking, believing, arguing, and sometimes making mistakes.



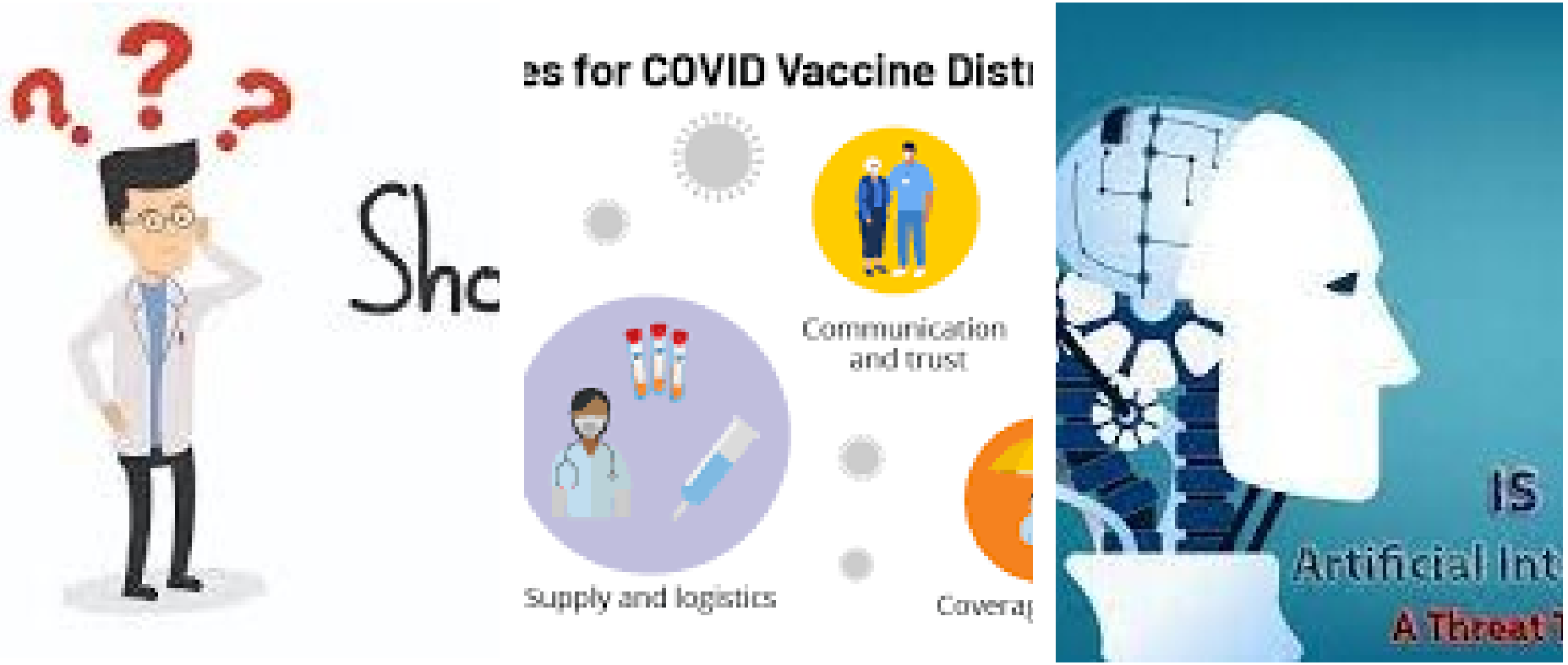
- Human beings can always decide what research is done, and what is done with the results
- Science and technology are not the product of some unstoppable force, but are human products which both shape, and are shaped by, the society from which they emerge.



- Science and technology has brought lots of benefits and opportunities to humankind, but there are increasing concerns over science and technology due to many reasons such as the use of S&T in WWII and Vietnam war



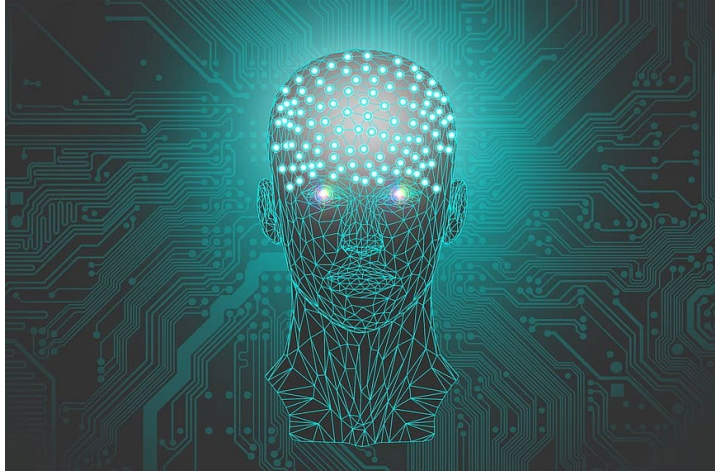


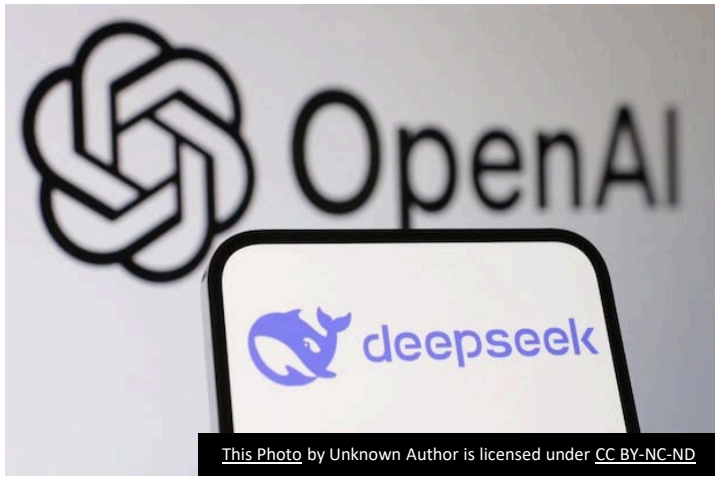


Issues in science and technology

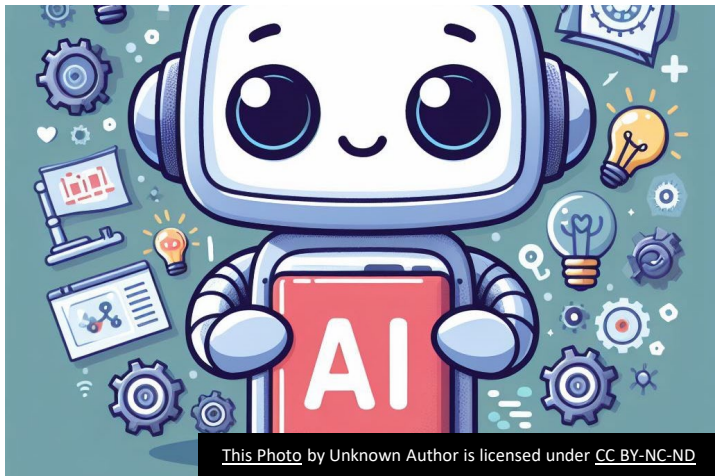
Issue 1 – Artificial Intelligence (AI) and Ethical Concerns

- AI is transforming fields such as healthcare, finance, and surveillance.
- Key concerns:
  - **Bias in AI algorithms** (e.g., racial or gender bias in hiring and policing systems)
  - **Privacy risks** (e.g., facial recognition and personal data misuse)
  - **Job displacement** due to automation
  - **Accountability and decision-making** (e.g., AI in law and healthcare)





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So..why should YOU learn STS?



To explore science and technology from different perspectives



To gain more understanding about influence of science and technology on society and vice versa



To learn issues in science and technology



Towards becoming a well-rounded, socially responsible professional in science or nonscience

STS for a sustainable future of humanity

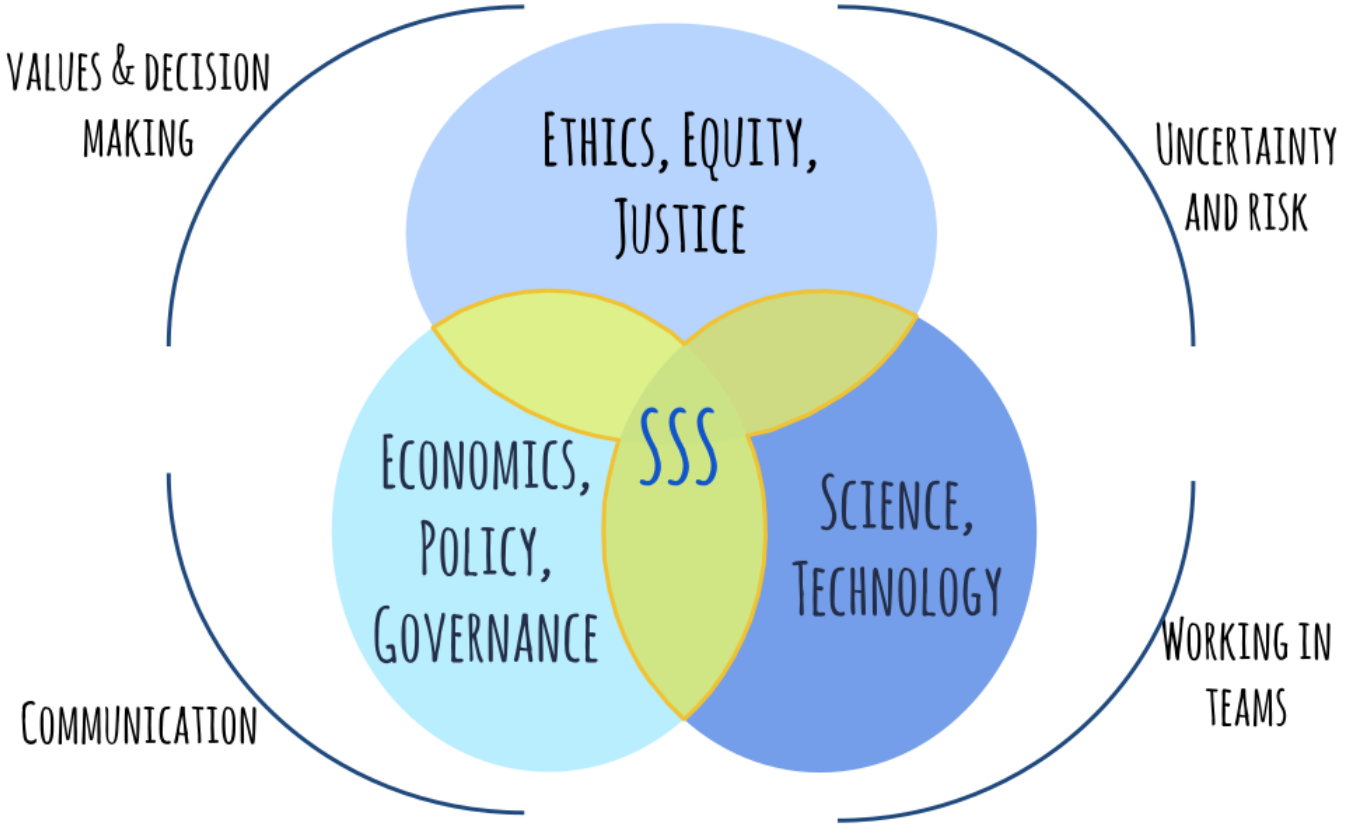


Figure: Key knowledge and skills to address sustainability challenges (https://www.mcgill.ca/sss/)

Quick reflection

- Three important ideas: **science, technology, and society**.
- **History and philosophy of science**,
- **Karl Popper** and **Thomas Kuhn**
- **technology**,
- **Science, Technology and Society, or STS**,

Main references

Bridgstock, M., Burch, D., Forge, J., Laurent, J., & Lowe, I. (1998). *Science, technology and society: an introduction*. Cambridge University Press.

Bucchi, M. (2004). *Science in society: An introduction to social studies of science*. Routledge.

Grant, E. (1997). HISTORY OF SCIENCE: When Did Modern Science Begin?. *The American Scholar*, *66*(1), 105-113.

McGinn, R. E. (1991). Science, Technology and Society. New Jersey: Prentice Hall.

Sismondo, S. (2004). An introduction to science and technology studies. Chichester: Wiley-Blackwell.- Chapter 1